

Inclusive Web Design: embracing diversity

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Source: Unsplash @marcobian

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User-Centred Design, as the name suggests, revolves around the user. It takes into account his **goals** and his **expectations**, as well as his **struggles**, both known and possibly discoverable. Every designer must carefully evaluate the group of people for whom he is designing and, to do so, research is fundamental.

However, while trying to identify the group of target users, we are at risk of encountering many possible mistakes: one of them is concentrating exclusively on common, typical or even “**normal**” **users**, while ignoring other categories that will be excluded by default.

This is a major disadvantage both for the user, who hasn't been considered during the design process, and for the client, since his product will have a limited audience. User Experience doesn't focus only on usability but also on the **pleasurability** of the design. So even if a user can practically and successfully use the product or service offered, we can't exclude the possibility that he may not enjoy it.

The aim of this paper is to address the concept of **inclusive design** online and offline, how it differs from accessible and universal design, how to recognize and prevent exclusion and, lastly, the importance of learning from human diversity.

Accessible Design and Universal Design

In order to fully understand what inclusive design is about, it's useful to introduce two related and popular concepts: **accessible design** and **universal design**. The main similarity between them is the intent to remove barriers that prevent a person from using a product or service.

Accessible Design

Accessible design aims to grant access to an interface, a technology, a product or a service by making sure it is usable by people regardless of their **disabilities**. From the web perspective, as the W3C defines it, accessibility addresses discriminatory aspects related to equivalent user experience for people with disabilities. Web accessibility means that people with disabilities

can equally perceive, understand, navigate, and interact with websites and tools. It also means that they can contribute equally without barriers¹.

Its focus is defined by guidelines or standards, such as the Web Content Accessibility Guidelines (WCAG), therefore its goals are measurable and easy to verify. As described later, accessibility is an **attribute**, while inclusive design is a **method**².

By practicing inclusive design, the product should become at least more accessible, but it's not necessarily the same the other way around. Accessibility doesn't consider the diversity of people beyond their difficulties. Ideally, accessibility and inclusive design work together to make experiences that are not only compliant with standards, but truly usable and open to all. The practice of accessibility fundamentals is a great way to begin to understand inclusivity.

What is a disability?

According to Cambridge Dictionary, a **disability** is “an illness, injury, or condition that makes it difficult for someone to do the things that other people do³”. It is therefore presented as a defect of the individual, which forces him into a disadvantageous position.

An interesting definition of disability is given by the **Social Model of Disability**⁴, which totally differs from the Medical Model. According to it, a disability is the **result of the interaction** between the physical, sensorial, cognitive or mental individual limitation and the context of life.

In other words, a person is disabled because the context makes him or her such: the barriers are built by the environment, not by the person. This definition puts society and, more specifically, the designer in a position of responsibility.



¹ W3C, “Accessibility, Usability, and Inclusion”. Link: <https://www.w3.org/WAI/fundamentals/accessibility-usability-inclusion/>

² Kat Holmes (2018), “Mismatched: How Inclusion Shapes Design”, The MIT Press Cambridge.

³ Cambridge Dictionary (online version), consulted on December 7th 2022.

⁴ Scope, Social model of disability. Link: <https://www.scope.org.uk/about-us/social-model-of-disability/>

Nobody would feel disabled if products and services were designed without barriers by default. On the same page, the World Health Organization (WHO) adds that “a person’s environment has a huge effect on the experience and extent of disability. Inaccessible environments create barriers that often hinder the full and effective participation of persons with disabilities in society on an equal basis with others.⁵”

A disability can be **permanent**, **temporary** or **situational**.

1. **Vision impairment:** blindness, low vision, color blindness.
2. **Motion impairment:** total or partial absence or immobility of limbs, slowness, shakiness and muscular pain.
3. **Hearing impairment:** of every level to complete deafness.
4. **Cognitive impairment:** dyslexia, attention, comprehension and memory difficulties.
5. **Speech impairment:** non-verbal.

Following the definition of temporary/situational impairment, we can imagine other disabilities caused by **context**.

1. **Vision impairment:** sudden flash, vertigo.
2. **Motion impairment:** broken arm or holding bags.
3. **Hearing impairment:** being in a loud place.
4. **Cognitive impairment:** foreign language.

	Permanent	Temporary	Situational
Touch	 One arm	 Arm injury	 New parent
See	 Blind	 Cataract	 Distracted driver
Hear	 Deaf	 Ear infection	 Bartender
Speak	 Non-verbal	 Laryngitis	 Heavy accent

Source: Microsoft Design - The Persona Spectrum

⁵ World Health Organization, “Disability: overview”. Link: https://www.who.int/health-topics/disability#tab=tab_1

5. **Speech impairment:** strong social anxiety and stuttering, drunkenness.

Guidelines for accessibility

In the **WCAG 2.1** (Web Content Accessibility Guidelines)⁶, written by the W3C (World Wide Web Consortium), are listed guidelines to design for accessibility. Here are some of them:

- **Ensure contrast is adequate:** a light background with a dark text is the basic combination to provide good legibility. Broadly speaking, the goal is a contrast ratio of 4.5:1, while a ratio of 3:1 is necessary when using fonts bigger than 18 pt. It's possible to test the design by visiting www.accessible-colors.com. This rule should also be applied to images, not only texts.
- **Use blank spaces:** leaving a blank space between different parts of the page is really helpful to differentiate them and to give a more organized appearance, which also helps in processing the information.
- **Be careful when selecting a font:** the font must be legible and of adequate size and spacing. Usually, the easier-to-read fonts are the simplest sans-serifs. Some of the most popular are Arial, Helvetica, Montserrat, Open Sans and Lato.
- **Take into account readability:** apart from legibility (font, contrast and spacing), readability must also be considered. It's about the content of the text and its ease of understanding. Therefore, it's recommended to employ clear, fluent and simple speech. This is essential for dyslexic users, also.
- **Incorporate transcriptions in pictures and videos:** audio and video products should always have a text alternative, as they can be useful to visually impaired people using screen readers. A transcription can also be useful to users with a short attention span or with difficulties in understanding the language.
- **Avoid strong light effects:** it is advisable to avoid designing products that can cause seizures due to light-sensitive epilepsy, like fast-blinking lights or fast-changing pictures. If they can't be avoided a trigger warning should be communicated.
- **Always give feedback:** the user must also be sure that his request is being processed. As such, it's important to use more than one sensory feedback, like a text and a sound.

⁶ W3C, Web Content Accessibility Guidelines 2.1. Link <https://www.w3.org/TR/WCAG21/>

- **Account for people using exclusively a keyboard:** not everyone is going to use both a mouse and a keyboard. For example, the selected item should be highlighted even if not selected through a mouse.
- **Do not use (only) color to mark, divide and differentiate items:** while browsing, it's better to highlight the displayed page by underscoring its name or making it bold. Using only colour is not enough to emphasize an item, like an error message. It's also advised to write it bigger and/or bold. This is necessary for colorblind people. Online there are many tools to verify if the chosen colour palette is suitable.
- **Attach written descriptions to pictures:** always add an alternative text to pictures, as some people use assistive technologies⁷, that can read the description added to an image. However, descriptions shouldn't be too long to prevent slowing the experience.

Inclusive Design

In the second episode of the first season of "Wireframe" podcast⁸ Khoi Vinh and Isabella Kulkarni tell us how inclusivity was born. The first TV show to use captions was Julia Child's, a chef that presented French dishes to Americans. Captions can be useful to many people: those with hearing impairment and foreigners, but also to those who simply like to read and turn off the volume. However, captions were designed specifically for the audience with hearing impairment.

The use of open-captions was not liked by everyone: they were not removable so even those without hearing problems were forced to see them. Many found them annoying and limiting, as they were covering the scene. The mistake was that while trying to find an accessible solution for those with a disability, the designers forgot about those that don't have it or simply won't like it. In trying to remove a limit, they created one. The issue was later solved with closed-captions: **optional captions that the user could active whenever he wanted.**

Design for diversity, not for limits

⁷ Assistive technologies are IT tools used to grant disabled people access and use to hardware and software. They are interfaces that can be installed on a standard system, like a computer or a smartphone, to make it accessible. Some examples include: personalized keyboards, pressure sensors, screen readers and ergonomic mouses.

⁸ Podcast "Wireframe" by Adobe: *"Inclusivity is a recipe for Good Design"*, Season 1, Ep. 2, published on November 20th 2018, presented by Khoi Vinh and Isabella Kulkarni. Link: <https://xd.adobe.com/ideas/perspectives/wireframe-podcast/inclusivity-is-a-recipe-for-good-design-episode-7/>

A similar recent example is the diffusion of **audiobooks**. They help people with visual or motion disabilities and at the same time they help those who want to read while doing something else too.

In her TED Talk, Elise Roy explains that she gradually lost the ability to hear and that gave her a new perspective on the design of things⁹.



She asked herself: “What if we start to design for disabilities first? When we do so, we find new solutions that are inclusive and most of the times even better.”

💡 **Solve for one, extend to many**

Microsoft Design provides a complete description of **Inclusive Design**, as it states that it "is a methodology, born out of digital environments, that enables and draws on the full range of **human diversity**. Most importantly, this means including and learning from people with a range of perspectives"¹⁰.

Human diversity can include physical and cognitive ability, age, culture, economic position, education level, gender, place, language and race. Inclusive design is a design philosophy that accommodate the way people are: different boundaries, different abilities and ways to be at ease.

It means going farther the removal of obstacles and creating a better experience: **laying down a red carpet**. Inclusive design acknowledges that solutions that work for people with disabilities can benefit people in other

⁹ Youtube, Ted Talk: “*When we design for disability, we all benefit*”, posted on August 16th 2016, presented by Elise Roy. Link: <https://www.youtube.com/watch?v=g2m97gPI70I>

¹⁰ Microsoft Design, “*Inclusive 101*”. Link: <https://www.microsoft.com/design/inclusive/>. Downloaded in December 2022.

circumstances and that it's crucial to offer same opportunities and multiple ways to experience. The key work for a designer that applies the inclusive methodology is **empathy**¹¹.

As Susan Goltsman beautifully said: "Inclusive design doesn't mean you're designing one thing for all people. You're designing a diversity of ways to participate so that everyone has a **sense of belonging**"¹². It enables everyone to participate equally, with confidence and independence in everyday activities and creates new opportunities to deploy creative and problem-solving skills.



💡 **Solve for one, extend to many**

(More) guidelines for inclusive web design

The previous accessibility guidelines are also applied in an inclusive perspective. We may also add the followings:

- **Provide a dark mode option:** consider, but do not force, a dark mode since it can be appreciated by users that suffer from eyesight fatigue while it is more comfortable in dark spaces and facilitate battery saving.
- **Offer multiple languages:** do not assume that the user knows a specific language, which can be different from his location.

¹¹ YouTube, NN/g: "Inclusive Design is a Welcome Mat", posted on June 21st 2019. Link: <https://www.youtube.com/watch?v=2pBILUE49hl>

¹² Microsoft Design, "Inclusive: The film". Link <https://www.microsoft.com/design/inclusive/>

- **Include different demographic options** (for race, gender, and sexual orientation). For example, Tinder offers more than 30 options to define gender identity. Allowing the user to choose what information to give and make him describe himself on his own terms is an inclusive practice. It can be already a great help to offer multiple selections (checkboxes) instead of radio buttons.
- **Use inclusive illustrations**, to visually depict diversity. A common example is the variety of face emojis offered by Whatsapp, initially only yellow and now with multiple skin tones and hair types.

Universal Design

Sometimes it's materially impossible to offer more than one solution: costs, management, space and other resources make it harder to be inclusive. Think about your local **playground**: there are one or two swings, a slide, a few benches and other games to climb and jump.

As an adult, I sadly have observed multiple times how swings are not meant to be used by me: they are too close to the ground and I can barely gain speed while playing. A customizable swing would be an inclusive option: chains that can be shortened or elongated according to the height of the user and multiple types of seats with different sizes.

However, that's not economically and practically feasible, so a sacrifice has to be made: prioritize basic accessibility. In this way we will have now swing for kids and another with a belt for babies. This is a loose example of **universal design**, with an intent of accessibility.



Source: *Eastern Jungle Gym*

Even though universal design is often confused with inclusive design, it is definitely more ambitious. Its goal is to create a unique experience shared by the highest number of people, ideally everyone. A **single solution** that does not require to be adapted, therefore it is mostly rooted in architecture and environmental design.

The **Disability Act 2005**¹³ defines Universal Design:

- (a) means the design and composition of an environment so that it may be accessed, understood and used— (i) to the greatest practicable extent, (ii) in the most independent and natural manner possible, (iii) in the widest possible range of situations, and (iv) without the need for adaptation, modification, assistive devices or specialised solutions, by persons of any age or size or having any particular physical, sensory, mental health or intellectual ability or disability, and
- (b) means, in relation to electronic systems, any electronics-based process of creating products, services or systems so that they may be used by any person.

Universal design is strongest at describing the qualities of a final design, since it precisely describes the nature of physical objects. Inclusive design focuses on how a designer arrived at that solution, prioritizing the involvement of communities.

In a nutshell

Summarizing, inclusive design might not necessarily lead to universal designs and universal designs might not involve the participation of excluded communities. Accessible solutions aren't always designed to consider human diversity or emotional qualities like beauty or dignity, since they simply need to provide access.

¹³ Disability Act n. 14 (2005), Part 6, Chapter IA, pp. 45-46. Link: <https://www.wicklow.ie/Portals/0/Documents/Governance/Access-Disability/The-Disability-Act-2005/Disability%20Act%202005.pdf>

APPROACH	IN A SENTENCE	DIFF.	CONTEXT
ACCESSIBILITY DESIGN	No physical barriers	★	Both physical and digital
INCLUSIVE DESIGN	One-size-fits-one	★★	Digital (it's easy and efficient to adapt interfaces)
UNIVERSAL DESIGN	One-size-fits-all	★★★★	Mostly physical (creating more variations would be too expensive)

Design can have a great impact. It can help reach a wider audience, extend the market, as well as have great consequences on society by offering the same **opportunities**. Inclusive design also reflects how people really are and helps with the process of normalization of diversity.

It's important as designers to use our creativity and lateral thinking to find innovative and individual solutions, designing for real people in all their variability and getting the best out of the resources available.

Recognizing exclusion

Mismatches, as Kat Holmes said, are the building blocks of exclusion¹⁴. They can feel like little moments of exasperation when a technology product doesn't work the way we think it should, or they can feel like running into a locked door marked with a big sign that says "keep out."

However the author claims that exclusion isn't inherently bad, nor inclusion inherently good: **barriers to equality are problematic**. These factors are amplified by the digital age, where the modern marketer, engineer or designer is expected to build solutions that reach millions or even billions of people. At that scale, one small exclusionary misstep can have an amplifying negative effect. By contrast, one small change toward inclusion can benefit many people in a positive way.

Designing for inclusion starts with recognizing exclusion

When designers do research with users they tend to focus on the discovery of similarities and patterns, useful to create user personas. This process goes against what inclusive design is about. Interestingly, human-to-human interactions are full of mismatches, but people are able to adapt themselves in an effort to connect with each other¹⁵.

¹⁴ Kat Holmes (2018), *"Mismatched: How Inclusion Shapes Design"*, The MIT Press Cambridge.

¹⁵ *Ibidem*

The first mistake is made while defying the **user personas**. Their creation is a valid tool that helps to focus on the target of the product or service throughout its design, but they can also be a dangerous element of shallowness. The paradox is that through the user persona we are forced into a simplified view of complex, different and diverse beings: people.

Piotr Żrołka, accessibility expert, suggests we change the way we see our users. User personas obstruct our view of the most important facets of people (needs, limits, desires) with sterile data, like demographics. The solution is to not linger on appearances, but to work hard in the research phase¹⁶. Therefore, it's crucial to include people in the process, allowing the creation of innovative and creative solutions, as they can provide the designer with a wide range of perspectives.

Inclusive design puts people in the centre from the very start of the process, and those fresh, diverse perspectives are the key to true insight. Microsoft Design suggests using the Persona Spectrum as a tool to help foster empathy. When designing, we should ask ourselves “who are we excluding?” and “what abilities are necessary to use our design?”. These questions can help stimulate empathy and reflection.

Moreover, empathy is built through listening. A designer should be very careful not to fall into **biases**. Instead, they should always be brave enough to go against themselves, question their decisions and avoid being over-confident when research does not support the design decisions made.

Lastly, another important strategy is to always try to offer alternatives to the user and let him explore and experience the product or service freely (or at least in a personalized way).

Barriers to inclusion

Kat Holmes identified three main reasons why inclusion is hard to be applied¹⁷:

1. **Inclusion isn't nice**: one of the most common fears related to inclusion is a fear of using the wrong words, look bad or offend someone.
2. **Inclusion is imperfect**: a common concern of designers is being forced to create a lowest-common-denominator design. Trying to please everyone is good for no one.

¹⁶ Webinar: "11 ways to start your inclusive, accessible design toolkit" by UXPin, with Piotr Żrołka, aired on January 28th 2021. Link: <https://www.youtube.com/watch?v=1-IJzpTJYMU>

¹⁷ Kat Holmes (2018), *Mismatched: How Inclusion Shapes Design*, The MIT Press Cambridge.

3. **Inclusion is ongoing:** urgency, especially in growth-led businesses, is a constant pressure. It might not be possible to definitively design this elusive thing we call inclusion. Exclusion, conversely, is recognizable. It's measurable and tangible; when someone is excluded they know it unequivocally.

Learning from human diversity

**“Always remember that you are absolutely unique.
Just like everyone else.”**

– Margaret Mead, anthropologist

A good approach to analyze the relationship between people, activity, context and technology to design a good interface is to apply **PACT**¹⁸. It's usually the result of a group brainstorming, with the objective of identifying the needs for the designing of interactive systems.

People do activities in a specific context with technologies:

1. **People:** we must think of physical differences (height, weight, disabilities,...), psychological ones (language, memory, desires,...), social ones (motivations, hobbies,...) and mental models (built on expectations and experiences);
2. **Activities** that people want to do: there are many attributes: time (regularity and duration), complexity, safety, pressure, continuity and the nature of the practice;
3. **Context** where the activities happen: it includes the material place (location, noise, crowds,...), the social context and the organizational one;
4. **Technology:** the hardware and software components. We must consider input, output and communication.

Learning through **research** and **conversation** are the best ways to start an inclusive approach. When testing and evaluating interfaces, it's important to ensure a **diverse representation** of users and target their real needs. Educate

¹⁸ UX Planet, “*Usability for Designers: P-A-C-T Framework*”, written by Muditha Batagoda, on January 27th 2018. Link: <https://uxplanet.org/usability-for-designers-p-a-c-t-framework-20509afc57>

stakeholders too on how different people use technology and involve them in your research, share examples and highlight user quotes.

It's easy to fall in the false-consensus effect¹⁹, defined in 1977 by Ross, Greene, and House, which refers to people's tendency to assume that others share their beliefs and will behave similarly in a given context. Only people who are very different from them would make different choices.

 **You are not the user**

Conclusions

Thinking of people instead of persona and of the context where the product or system will be used, together with the way it will be employed, can be useful to assess new possibilities and discover obstacles not considered before.

A few key points:

- Recognize exclusion. Identify ability biases and mismatched interactions that are related to your solution.
- Don't be biased into thinking your product is made for everyone.
- Aim to reflect how people really are, not an oversimplified version.
- Create a diversity of ways to participate in an experience.

An inclusive mentality should not be seen as a limit to creativity, but as a challenge to reconsider and stimulate new ideas and perspective.

Designing for inclusivity not only opens up our products and experiences to more people with a wider range of abilities. It also reflects how people really are. All humans are growing, changing, and adapting to the world around them every day. We want our designs to reflect that diversity. Every decision we make can raise or lower barriers to participation in society²⁰.

¹⁹ Nielsen Norman Group, "*You Are Not The User: The False-Consensus Effect*" written by Raluca Budiú on October 22, 2017. Link: <https://www.nngroup.com/articles/false-consensus/>

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